

# Selected solutions to Atiyah-Macdonald's exercises

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## Abstract

A non-comprehensive collection of solutions to Atiyah-Macdonald's exercises, with some additional comments and discussions. There may be errors, for reference only.

## 1 Chapter 2

**Exercise 1** ([AM69]-exr-2.1).  $\mathbb{Z}/m\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} = 0$  when  $m, n$  are coprime.

*Proof.* By the Chinese Remainder Theorem, there exists integers  $a, b$  such that

$$am + bn = 1$$

Then for any  $x \otimes y \in \mathbb{Z}/m\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z}$ ,

$$x \otimes y = 1 \cdot (x \otimes y) = (am + bn)(x \otimes y) = am(x \otimes y) + bn(x \otimes y) = 0 + 0 = 0.$$

Thus  $\mathbb{Z}/m\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} = 0$ . □

**Exercise 2** ([AM69]-exr-2.3). For  $A$  a local ring,  $M, N$  finitely generated  $A$ -modules. If  $M \otimes_A N = 0$ , then  $M = 0$  or  $N = 0$ .

*Proof.* Tensoring with the residue field  $k := A/\mathfrak{m}$  (where  $\mathfrak{m}$  is the maximal ideal of  $A$ ), we have

$$(M \otimes_A k) \otimes_k (N \otimes_A k) = 0$$

Since it's a tensor product of vector spaces over field  $k$ , we must have  $M \otimes_A k = 0$  or  $N \otimes_A k = 0$ , by dimension counting. Without loss of generality, say  $M \otimes_A k = 0$ . By Nakayama's lemma, this implies  $M = 0$ . □

**Exercise 3** ([AM69]-exr-2.11). Let  $A$  be a nonzero ring. Show that  $A^m \cong A^n$  iff  $m = n$ .

Additionally, show that

- if  $\varphi : A^m \rightarrow A^n$  is surjective, then  $m \geq n$ .
- If  $\varphi$  is injective, is it true that  $m \leq n$ ?

*Proof.* By taking tensor product with  $k := A/\mathfrak{m}$  (where  $\mathfrak{m}$  is any maximal ideal of  $A$ , existence by non-zeroness) and counting dimensions as vector spaces. The first additional question by the right-exactness of tensor product. The second question seems rather hard [TODO]. □

**Exercise 4** ([AM69]-exr-2.14: Direct limits). Definition of direct limit of a direct system of  $A$ -modules  $(M_i, \mu_{ij})_{i,j \in I}$  indexed by a directed set  $I$ .  $M := C/D$  where  $C$  is the direct sum and  $D$  is the relations.

*Proof.* There's nothing to prove. □

**Exercise 5** ([AM69]-exr-2.15). Every element of  $M := \varinjlim_i M_i$  is of the form  $\mu_i(x_i)$  for some  $i \in I$  and  $x_i \in M_i$ .

If  $\mu_i(x_i) = 0$  in  $M$ , then  $\exists j \geq i$  s.t.  $\mu_{ij}(x_i) = 0$  in  $M_j$ .

*Proof.* Say  $M \ni x = \sum_i \mu_i(x_i)$ , where  $i$  runs over a finite subset of  $I$  and  $x_i \in M_i$ . By directedness of  $I$ ,  $\exists k \in I$  s.t.  $k \geq i$  for all  $i$  in the finite subset. Then  $\mu_k(\sum_i \mu_{ik}(x_i)) = x$ .

If  $\mu_i(x_i) = 0$  in  $M$ , then  $x_i \in M_i \cap D$ , that is, a finite sum in  $C$

$$x_i = \sum_{j < k} y_{jk} - \mu_{jk}(y_{jk}) = \sum_k \left( \sum_{k < l} y_{kl} - \sum_{j < k} \mu_{jk}(y_{jk}) \right)$$

Comparing the summation componentwise in  $C$  gives

$$\begin{aligned} x_i &= \sum_{i < l} y_{il} - \sum_{j < i} \mu_{ji}(y_{ji}) \\ 0 &= \sum_{k < l} y_{kl} - \sum_{j < k} \mu_{jk}(y_{jk}) \quad \text{for all } k \neq i \end{aligned}$$

Let  $p$  be an upper bound of all indices appearing in the above equations. Apply  $\mu_{ip}$  to the first equation and  $\mu_{kp}$  to the second equation for all  $k \neq i$ . We get

$$\begin{aligned} \mu_{ip}(x_i) &= \sum_{i < l} \mu_{ip}(y_{il}) - \sum_{j < i} \mu_{jp}(y_{ji}) \\ 0 &= \sum_{k < l} \mu_{kp}(y_{kl}) - \sum_{j < k} \mu_{jp}(y_{jk}) \quad \text{for all } k \neq i \end{aligned}$$

Summing up all these equations gives  $\mu_{ip}(x_i) = 0$ . □

**Exercise 6** ([AM69]-exr-2.16). The universal property of direct limits.

*Proof.* “Up to isomorphism” part by abstract nonsense of category theory.

Say we have  $\alpha_i : M_i \rightarrow N$  s.t.  $\alpha_j \circ \mu_{ij} = \alpha_i$  for all  $i \leq j$ . Define  $\varphi : C \rightarrow N$  by  $\varphi \circ \mu_i := \alpha_i$ . By the universal property of direct sum, such  $\varphi$  is well-defined. Also note that this is the only possible definition of  $\varphi$  satisfying  $\varphi \circ \mu_i = \alpha_i$ . This uniqueness also passes down to the quotient  $M = C/D$ . Say the quotient map is  $\bar{\varphi} : M = C/D \rightarrow N$ , it remains to show that  $\bar{\varphi}$  is well-defined. For any generator of  $D$ , say  $x_j - \mu_{jk}(x_j)$  for some  $j \leq k$  and  $x_j \in M_j$ , we have  $\varphi(x_j - \mu_{jk}(x_j)) = \alpha_j(x_j) - \alpha_k(\mu_{jk}(x_j)) = 0$ . □

**Exercise 7** ([AM69]-exr-2.17). Any  $A$ -module is a direct limit of its finitely generated  $A$ -submodules.

*Proof.* It's clear that  $\sum_i M_i = \bigcup_i M_i$  when the direct system is directed by inclusion. It's also easy to verify that  $\bigcup_i M_i$  satisfies the universal property of direct limits. □

**Exercise 8** ([AM69]-exr-2.18).  $\mathbf{M} := (M_i, \mu_{ij})$ ,  $\mathbf{N} := (N_i, \nu_{ij})$  direct systems of  $A$ -modules indexed by the same directed set  $I$ . Let  $M, N$  be their direct limits and  $\mu_i, \nu_i$  be the canonical maps, respectively.

A homomorphism  $\Phi : \mathbf{M} \rightarrow \mathbf{N}$  is defined to be a family of homomorphisms  $\varphi_i : M_i \rightarrow N_i$  satisfying  $\nu_{ij} \circ \varphi_i = \varphi_j \circ \mu_{ij}$  for all  $i \leq j$ . Show that  $\Phi$  defines a unique homomorphism  $\varphi = \varinjlim \Phi : M \rightarrow N$  such that  $\varphi \circ \mu_i = \nu_i \circ \varphi_i$  for all  $i$ .

That is, taking direct limits is a functor from the category of direct systems of  $A$ -modules to the category of  $A$ -modules.

*Proof.* Let  $\bar{\varphi}_i$  be the composition  $\nu_i \circ \varphi_i : M_i \rightarrow N_i \rightarrow N$ . Define our map  $\varphi : M \rightarrow N$  by these maps, the required compatibility condition  $\varphi \circ \mu_i = \nu_i \circ \varphi_i = \bar{\varphi}_i$  is guaranteed by hypothesis.  $\square$

**Exercise 9** ([AM69]-exr-2.19). Say  $\mathbf{M} := (M_i)_{i \in I}$ ,  $\mathbf{N} := (N_i)_{i \in I}$ ,  $\mathbf{P} := (P_i)_{i \in I}$  are direct systems of  $A$ -modules indexed by the same directed set  $I$ . If the sequence of direct systems

$$\mathbf{M} \longrightarrow \mathbf{N} \longrightarrow \mathbf{P}$$

is exact, then the sequence of direct limits

$$M \longrightarrow N \longrightarrow P$$

is also exact. That is, taking direct limits is an exact functor from the category of direct systems of  $A$ -modules to the category of  $A$ -modules.

*Proof.* Say the maps in the direct systems are  $\varphi_i : M_i \rightarrow N_i$  and  $\psi_i : N_i \rightarrow P_i$ . The maps in the direct limits are  $\varphi : M \rightarrow N$  and  $\psi : N \rightarrow P$  induced by  $\varphi_i$  and  $\psi_i$  respectively. The canonical maps are  $\mu_i : M_i \rightarrow M$ ,  $\nu_i : N_i \rightarrow N$  and  $\pi_i : P_i \rightarrow P$ . These maps make the following diagram commute:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & M_i & \xrightarrow{\varphi_i} & N_i & \xrightarrow{\psi_i} & P_i & \longrightarrow & 0 \\ & & \downarrow \mu_i & & \downarrow \nu_i & & \downarrow \pi_i & & \\ 0 & \longrightarrow & M & \xrightarrow{\varphi} & N & \xrightarrow{\psi} & P & \longrightarrow & 0 \end{array}$$

Figure 1

We do diagram chasing. First, We have  $(\psi \circ \varphi) \circ \mu_i = \psi \circ \nu_i \circ \varphi_i = (\pi_i \circ \psi_i) \circ \varphi_i = 0$ . By the universal property of direct limits,  $\psi \circ \varphi = 0$ . Second, for any  $y \in N$  with  $\psi(y) = 0$ , say  $y = \nu_i(y_i)$  for some  $i \in I$  and  $y_i \in N_i$  (by Exercise 5). Then by the exactness at  $N_i$ , there exists  $x_i \in M_i$  s.t.  $\varphi_i(x_i) = y_i$ . Let  $x := \mu_i(x_i) \in M$ , then  $\varphi(x) = \varphi \circ \mu_i(x_i) = \nu_i \circ \varphi_i(x_i) = \nu_i(y_i) = y$ , hence the exactness at  $N$  follows.  $\square$

**Exercise 10** ([AM69]-exr-2.20). show that

$$\varinjlim_i (M_i \otimes N) \cong \left( \varinjlim_i M_i \right) \otimes N$$

i.e. tensor product commutes with direct limits.

*Proof.* It is known from category theory that a right exact functor commutes with colimits [TODO: prove this], but we write some concrete nonsense here anyway.

Suffices to show the universal property of direct limits holds for  $\left( \varinjlim_i M_i \right) \otimes N$  with respect to the direct system  $(M_i \otimes N, \mu_{ij} \otimes \text{id}_N)_{i,j \in I}$ .

First, determine the canonical inclusions. Define  $M_i \otimes N \rightarrow \left( \varinjlim_i M_i \right) \otimes N$  by  $\mu_i \otimes \text{id}_N$ , i.e. sending  $m_i \otimes n \mapsto \mu_i(m_i) \otimes n$  for  $m_i \in M_i$ ,  $n \in N$ .

Next, verify the universal property. Say we have maps  $\alpha_i : M_i \otimes N \rightarrow P$  satisfying the compatibility condition  $\alpha_i = \alpha_j \circ (\mu_{ij} \otimes \text{id}_N)$  for all  $i \leq j$ . To define a map  $\bar{\alpha}_i : \left( \varinjlim_i M_i \right) \otimes N \rightarrow P$ , it suffices to give a map  $\varinjlim_i M_i \rightarrow \text{Hom}_A(N, P)$ . This in turn require us to define maps  $M_i \rightarrow \text{Hom}_A(N, P)$ , and we define it as  $m_i \mapsto (n \mapsto \alpha_i(m_i \otimes n))$ . One verifies they commutes with transition maps  $\mu_{ij}$  thanks to the compatibility condition  $\alpha_i = \alpha_j \circ (\mu_{ij} \otimes \text{id}_N)$ ,

thus the map  $\varinjlim_i M_i \rightarrow \text{Hom}_A(N, P)$  is well-defined, and in turn  $\bar{\alpha}_i : \left(\varinjlim_i M_i\right) \otimes N \rightarrow P$  is defined.

Finally, note that this choice of definition satisfies, and is unique to satisfy the universal property condition  $\bar{\alpha}_i \circ (\mu_i \otimes \text{id}_N) = \alpha_i$ . Hence we have verified the universal property of  $\left(\varinjlim_i M_i\right) \otimes N$  (with the defined canonical inclusions) w.r.t. the direct system  $(M_i \otimes N, \mu_{ij} \otimes \text{id}_N)_{i,j \in I}$ .  $\square$

**Exercise 11** ([AM69]-exr-2.21).  $(A_i)_{i \in I}$  a family of rings indexed by a directed set  $I$ . For each  $i \leq j$ , let  $\alpha_{ij} : A_i \rightarrow A_j$  be a ring homomorphism such that  $\alpha_{ii} = \text{id}_{A_i}$  and  $\alpha_{jk} \circ \alpha_{ij} = \alpha_{ik}$  for all  $i \leq j \leq k$ . As  $\mathbb{Z}$ -module we may view them as a direct system and obtain their direct limit  $A := \varinjlim_i A_i$ . Show that  $A$  has a natural ring structure such that the canonical maps  $\alpha_i : A_i \rightarrow A$  are ring homomorphisms. This defines the direct limit in the category of rings.

In addition, show that  $A = 0$  implies  $A_i = 0$  for some  $i$ .

*Proof.* We define the multiplication on  $A$  as follows. For any  $x, y \in A$ , say  $x = \alpha_i(x_i)$  and  $y = \alpha_j(y_j)$  for some  $i, j \in I$  and  $x_i \in A_i, y_j \in A_j$  (by Exercise 5). Let  $k \in I$  be an upper bound of  $i, j$ . Define  $x \cdot y := \alpha_k(\alpha_{ik}(x_i) \cdot \alpha_{jk}(y_j))$ . We verify that this definition is independent of the choice of representatives  $x_i, y_j$  and the choice of upper bound  $k$ .

If we have another choice of representatives  $x_{i'} \in A_{i'}, y_{j'} \in A_{j'}$  for some  $i', j' \in I$  with  $x = \alpha_{i'}(x_{i'})$  and  $y = \alpha_{j'}(y_{j'})$ . Let  $k' \in I$  be an upper bound of  $i', j'$ . First we claim that there exist an  $l \in I$  s.t.  $\alpha_{il}(x_i) = \alpha_{il}(x_{i'})$ . Indeed, since  $\alpha_k(\alpha_{ik}(x_i)) = x = \alpha_k(\alpha_{i'k}(x_{i'}))$ , by Exercise 5 there exists an upper bound  $l$  of  $k, i'$  s.t.  $\alpha_{i'l}(x_{i'}) = \alpha_{il}(x_i)$ . Similarly, we can find an upper bound s.t.  $x_j$  and  $x_{j'}$  agree after applying the transition map to that upper bound. WLOG, still use the letter  $l$  be an upper bound of these two upper bounds. Then

$$\begin{aligned} \alpha_k(\alpha_{ik}(x_i) \cdot \alpha_{jk}(y_j)) &= \alpha_l(\alpha_{kl}(\alpha_{ik}(x_i) \cdot \alpha_{jk}(y_j))) \\ &= \alpha_l(\alpha_{il}(x_i) \cdot \alpha_{jl}(y_j)) \\ &= \alpha_l(\alpha_{i'l}(x_{i'}) \cdot \alpha_{j'l}(y_{j'})) \\ &= \alpha_{k'}(\alpha_{i'k'}(x_{i'}) \cdot \alpha_{j'k'}(y_{j'})) \end{aligned}$$

Thus the definition is independent of the choice of representatives and upper bounds.

$\alpha_i$  is a ring homomorphism is straightforward to verify.

if  $A = 0$ , then for any  $i \in I$ ,  $\alpha_i(1_{A_i}) = 0$  in  $A$ . By Exercise 5, there exists an upper bound  $j \in I$  of  $i$  s.t.  $\alpha_{ij}(1_{A_i}) = 0$  in  $A_j$ . But the ring homomorphism  $\alpha_{ij}$  must preserve identity, thus  $A_j = 0$ .  $\square$

**Exercise 12** ([AM69]-exr-2.22).  $(A_i, \alpha_{ij})$  direct system of rings. Let  $\mathfrak{R}_i$  be the nilradical of  $A_i$ . Show that  $\varinjlim_i \mathfrak{R}_i$  is the nilradical of  $\varinjlim_i A_i$ .

As a corollary, if each  $A_i$  is an integral domain, so is  $\varinjlim_i A_i$ .

*Proof.* For  $i \in I$ ,  $a_i \in A_i$ , TFAE:

- $\alpha_i(a_i) \in \mathfrak{R}(\varinjlim_j A_j)$
- Exists some  $n$  s.t.  $\alpha_i(a_i^n) = 0$
- Exists some  $n, j \geq i$  s.t.  $\alpha_{ij}(a_i^n) = 0$  (by Exercise 5)
- Exists some  $j \geq i$  s.t.  $\alpha_{ij}(a_i) \in \mathfrak{R}_j$
- $\alpha_i(a_i) \in \varinjlim_j \mathfrak{R}_j$

Thus  $\mathfrak{R}(\varinjlim_j A_j) = \varinjlim_j \mathfrak{R}_j$  by Exercise 5.  $\square$

**Exercise 13** ([AM69]-exr-2.23).  $(B_\lambda)_{\lambda \in \Lambda}$  a family of  $A$ -algebras. For each finite subset of  $\Lambda$ , let  $B_J$  be the tensor product of the  $B_\lambda$  for  $\lambda \in J$ . By inclusion of  $J$ , all  $B_J$  form a

direct system. Let  $B$  denote its direct limit. Then  $B$  has a natural  $A$ -algebra structure with  $B_J \rightarrow B$  are  $A$ -algebra homomorphisms. This  $B$  is called the tensor product of the family  $(B_\lambda)_{\lambda \in \Lambda}$  over  $A$ .

*Proof.* There is nothing to prove.  $\square$

**Exercise 14** ([AM69]-exr-2.24).  $M$  is flat iff  $\text{Tor}_i(M, N) = 0$  for all  $N$  iff  $\text{Tor}_1(M, N) = 0$  for all  $N$ .

$i \implies ii$ : Say

$$0 \rightarrow K \rightarrow P \rightarrow N$$

where  $P$  is projective. Write the long exact sequence:

$$\begin{array}{ccccccc} \text{Tor}_2(K, M) & \longrightarrow & \text{Tor}_2(P, M) & \longrightarrow & \text{Tor}_2(N, M) & \longrightarrow & \\ \text{Tor}_1(K, M) & \longrightarrow & \text{Tor}_1(P, M) & \longrightarrow & \text{Tor}_1(N, M) & \longrightarrow & \\ K \otimes M & \longrightarrow & P \otimes M & \longrightarrow & N \otimes M & \longrightarrow & 0 \end{array}$$

Since  $-\otimes M$  is an exact functor.  $K \otimes M \rightarrow P \otimes M$  is injective, i.e.  $\text{Tor}_1(N, M) \xrightarrow{0} K \otimes M$ . Meanwhile,  $\text{Tor}_1(P, N) = 0$  since  $P$  is projective. This forces  $\text{Tor}_1(N, M) = 0$ . Thus for all  $N$ ,  $\text{Tor}^1(N, M) = 0$ . In particular,  $\text{Tor}^1(K, M) = 0$ . With  $\text{Tor}_2(P, M) = 0$ , this forces  $\text{Tor}^2(N, M) = 0$ . Then inductively work upward gives  $\text{Tor}_i(N, M) = 0$  for all  $i \geq 1$ .  $\square$

$ii \implies iii$ . Trivial.  $\square$

$iii \implies i$ . The long exact sequence.  $\square$

**Exercise 15** ([AM69]-exr-2.25).

$$0 \longrightarrow N_1 \longrightarrow N_2 \longrightarrow N_3 \longrightarrow 0$$

is exact.  $N_3$  is flat. Then  $N_1$  is flat iff  $N_2$  is flat.

*Proof.* Write the long exact sequence:

$$\begin{array}{ccccccc} \text{Tor}_2(N_1, M) & \longrightarrow & \text{Tor}_2(N_2, M) & \longrightarrow & \text{Tor}_2(N_3, M) & \longrightarrow & \\ \text{Tor}_1(N_1, M) & \longrightarrow & \text{Tor}_1(N_2, M) & \longrightarrow & \text{Tor}_1(N_3, M) & \longrightarrow & \\ N_1 \otimes M & \longrightarrow & N_2 \otimes M & \longrightarrow & N_3 \otimes M & \longrightarrow & 0 \end{array}$$

$\text{Tor}_i(N_3, M) = 0$  by  $N_3$  is flat.

If  $N_1$  is flat. i.e.  $\text{Tor}_i(N_1, M) = 0$ , then this forces  $\text{Tor}_i(N_2, M) = 0$ . i.e.  $N_2$  is flat.

Similarly for the case that  $N_2$  is flat.  $\square$

**Exercise 16** ([AM69]-exr-2.26: Characterization of flatness via finitely generated ideals).  $N$  is an  $A$ -module. Then  $N$  is flat iff  $\text{Tor}_1(A/\mathfrak{a}, N) = 0$  for all finitely generated ideals  $\mathfrak{a}$  in  $A$ .

$\implies$ . Trivial (by Exercise 14).  $\square$

$\Leftarrow$ . The hypothesis is that for all finitely generated ideals  $\mathfrak{a}$  of  $A$ , the sequence

$$0 \longrightarrow \mathfrak{a} \otimes N \longrightarrow A \otimes N \longrightarrow A/\mathfrak{a} \otimes N \longrightarrow 0$$

is exact.

We first prove that for any ideal  $\mathfrak{a}$  we have  $\text{Tor}_1(A/\mathfrak{a}, N) = 0$ . Consider the directed system of all finitely generated subideals of  $\mathfrak{a}$ , say  $(\mathfrak{a}_i)_{i \in I}$ . Its direct limit is  $\mathfrak{a}$ . Then we have an exact sequences of direct systems:

$$0 \longrightarrow (\mathfrak{a}_i)_{i \in I} \otimes N \longrightarrow \mathbf{A} \otimes N \longrightarrow (A/\mathfrak{a}_i)_{i \in I} \otimes N \longrightarrow 0$$

Taking direct limits, we get an exact sequence

$$0 \longrightarrow \mathfrak{a} \otimes N \longrightarrow A \otimes N \longrightarrow A/\mathfrak{a} \otimes N \longrightarrow 0$$

(by Exercise 9 exactness of taking direct limits, Exercise 10 commutativity of direct limit and tensor product). This shows that  $\text{Tor}_1(A/\mathfrak{a}, N) = 0$  for any ideal  $\mathfrak{a}$ . i.e.  $\text{Tor}_1(A/\mathfrak{a}, N) = 0$ .

Note any cyclic  $A$ -modules are naturally isomorphic to  $A/\mathfrak{a}$  for some ideal  $\mathfrak{a}$  of  $A$ , thus  $\text{Tor}_1(M, N) = 0$  for any cyclic  $A$ -module  $M$ .

Next we show that for any finitely generated  $A$ -module  $M$ ,  $\text{Tor}_1(M, N) = 0$ . Consider a filtration of  $M$ :

$$0 = M_0 \subset M_1 \subset \cdots \subset M_m = M$$

such that each successive quotient  $M_i/M_{i-1}$  is cyclic. Such a filtration exists by taking generators of  $M$  step by step. Consider the short exact sequence

$$0 \longrightarrow M_{i-1} \longrightarrow M_i \longrightarrow M_i/M_{i-1} \longrightarrow 0$$

for each  $1 \leq i \leq m$ . Since  $M_i/M_{i-1}$  is cyclic, we have  $\text{Tor}_1(M_i/M_{i-1}, N) = 0$ . By the proof of Exercise 15, we have  $\text{Tor}_1(M_i, N) \cong \text{Tor}_1(M_{i-1}, N)$ . Thus

$$\text{Tor}_1(M, N) \cong \text{Tor}_1(M_{m-1}, N) \cong \cdots \cong \text{Tor}_1(M_1, N) \cong \text{Tor}_1(M_0, N) = 0.$$

Finally, for any  $A$ -module  $M$ , we can write it as a direct limit of its finitely generated submodules, say  $(M_i)_{i \in I}$ . Then by a similar argument as above using Exercise 9 and Exercise 10 applied to the exact sequences

$$0 \longrightarrow \text{Tor}_1(M_i, N) \longrightarrow K_i \otimes N \longrightarrow P_i \otimes N \longrightarrow M_i \otimes N \longrightarrow 0$$

where  $P_i$  is a free module, we know that for all  $A$ -module  $M$ ,  $\text{Tor}_1(M, N) = 0$  and hence  $N$  is flat.  $\square$

*Remark.* The last step essentially shows that the Tor functor commutes with direct limits, i.e. for a directed system of  $A$ -modules  $(M_i)_{i \in I}$ ,

$$\text{Tor}_n \left( \varinjlim_i M_i, N \right) \cong \varinjlim_i \text{Tor}_n(M_i, N)$$

for all  $n \geq 0$ . In particular,  $n = 0$  is exactly the result of Exercise 10. [TODO] The proof above is just a vague sketch for now.

## 1.1 Extra explorations on Tor and flatness

### The etymology of Tor: Quotient perspective

Say  $A$  is an integral domain and  $\mathfrak{a}$  is a principal ideal generated by  $a \in A$ . Then from the short exact sequence

$$0 \longrightarrow A \xrightarrow{a \cdot -} A \longrightarrow A/\mathfrak{a} \longrightarrow 0$$

tensoring  $N$  we obtain the long exact sequence (leftmost item is zero since  $A$  is projective)

$$0 \longrightarrow \text{Tor}_1(A/\mathfrak{a}, N) \longrightarrow N \xrightarrow{a \cdot -} N \longrightarrow N/\mathfrak{a}N \longrightarrow 0$$

We see that  $\text{Tor}_1(A/\mathfrak{a}, N) \cong \ker(a \cdot -) =: N[a]$ , the  $a$ -torsion submodule of  $N$ . This explains the etymology of Tor.

## The etymology of Tor: Localization perspective

Wikipedia if you want a quick reference for this part.

Another (somehow dual) way of understanding Tor is via localization. Say  $A_S$  is any localization of  $A$  with respect to a multiplicative set  $S$ . Consider the short exact sequence

$$0 \longrightarrow A \longrightarrow A_S \longrightarrow A_S/A \longrightarrow 0$$

Tensoring  $N$  to obtain the long exact sequence (leftmost item is zero since  $A_S$  as a localization of  $A$  is flat)

$$0 \longrightarrow \mathrm{Tor}_1(A_S/A, N) \longrightarrow N \longrightarrow S^{-1}N \longrightarrow (A_S/A) \otimes N \longrightarrow 0$$

We see that  $\mathrm{Tor}_1(A_S/A, N) \cong \ker(N \rightarrow S^{-1}N)$ . Thus  $\mathrm{Tor}_1(A_S/A, N)$  measures those elements in  $N$  that collapse to zero upon localization  $N \rightarrow S^{-1}N$ . What are these elements? When  $A$  is an integral domain, they are precisely those elements  $x \in N$  such that there exists  $s \in S$  with  $s \cdot x = 0$ , i.e. the  $S$ -torsion submodule of  $N$ .

In particular, if we take the localization to be the field of fractions  $k$ , then we have realized the torsion submodule  $T(M)$  of  $M$  as  $\mathrm{Tor}_1(k/A, M)$ , or the kernel of the localization map  $M \rightarrow k \otimes M$ . This gives another (probably better) explanation of the etymology of Tor.

Another example is to consider the localization at a prime ideal  $\mathfrak{p}$ , i.e.  $S = A \setminus \mathfrak{p}$ . In this case the terminology goes a little subtle... [TODO] continue this thought on PID.

It is worth noting that the kernel of  $k \otimes N \rightarrow (k/A) \otimes N$  captures the torsion-free part of  $N$ :

$$\ker(k \otimes N \rightarrow (k/A) \otimes N) \cong \mathrm{im}(N \rightarrow k \otimes N) \cong N / \ker(N \rightarrow k \otimes N) \cong N / T(N)$$

[TODO] continue this thought.

## Characterizing flatness

We continue our discussion in Section 1.1. Recall Exercise 16 suggests that, to characterize the flatness of  $A$ -module  $N$ , it suffices to understand  $\mathrm{Tor}_1(A/\mathfrak{a}, N)$  for all finitely generated ideals  $\mathfrak{a}$  of  $A$ . So we ask: If  $A$  is not in general an integral domain, and  $\mathfrak{a}$  is not in general principal, what is the meaning of  $\mathrm{Tor}_1(A/\mathfrak{a}, N)$ ?

In this case, we can still use the short exact sequence

$$0 \longrightarrow \mathfrak{a} \longrightarrow A \longrightarrow A/\mathfrak{a} \longrightarrow 0$$

to obtain the long exact sequence

$$0 \longrightarrow \mathrm{Tor}_1(A/\mathfrak{a}, N) \longrightarrow \mathfrak{a} \otimes N \xrightarrow{\quad \cdot \quad} N \longrightarrow A/\mathfrak{a} \otimes N \longrightarrow 0$$

From this we see that  $\mathrm{Tor}_1(A/\mathfrak{a}, N) \cong \ker(\mathfrak{a} \otimes N \xrightarrow{\quad \cdot \quad} N)$ , which can be viewed as the module of relations among the elements of  $\mathfrak{a}$  when they act on  $N$  via the  $A$ -scalar multiplication. This can be interpreted as somehow a generalization of the torsion elements in the principal ideal case.

Combine this with Exercise 16, one may say that the flatness of  $A$ -module  $N$  is precisely characterized by the fact that any finite subset  $\{a_1, \dots, a_n\} \subseteq A$  acts on  $N$  without introducing any “ $\mathfrak{a}$ -torsion” or “ $\mathfrak{a}$ -relations”, where  $\mathfrak{a}$  is the ideal generated by  $\{a_1, \dots, a_n\}$ . This intuition can be extremely useful, as the examples below will show.

**Example 17** (Pure ideals).

- For  $n \geq 2$ ,  $\mathbb{Z}/n\mathbb{Z}$  is not a flat  $\mathbb{Z}$ -module because it is not torsion-free.

Recall that a finitely generated module over a PID is free iff it is torsion-free, and in particular flat.

- $2\mathbb{Z}/6\mathbb{Z}$  is a flat  $\mathbb{Z}/6\mathbb{Z}$ -module, not only by that it is projective (a direct summand of  $\mathbb{Z}/6\mathbb{Z}$ ), but also because
  - 2 won't annihilate anything in  $2\mathbb{Z}/6\mathbb{Z}$ .
  - Despite that 3 annihilates  $2\mathbb{Z}/6\mathbb{Z}$ , we have  $3\mathbb{Z}/6\mathbb{Z} \otimes_{\mathbb{Z}/6\mathbb{Z}} 2\mathbb{Z}/6\mathbb{Z} = 0$ . For example,  $3 \otimes 2 = 3 \otimes 8 = 6 \otimes 4 = 0$ . Thus 3-torsion won't give any nontrivial relation in this case.

Blame  $\mathbb{Z}/6\mathbb{Z}$  for this subtlety! Recall that over an integral domain, any flat module is torsion-free. But  $\mathbb{Z}/6\mathbb{Z}$  is not an integral domain. In fact, any direct product of nontrivial rings ruins this property.

- Say  $k$  is a field,  $A := k[x, y]$ ,  $\mathfrak{a}$  is any nontrivial ideal of  $A$ . Then  $A/\mathfrak{a}$  is not a flat  $A$ -module.

In fact, pick any nonzero  $a \in \mathfrak{a}$ , then  $a \otimes 1 \in \mathfrak{a} \otimes_A A/\mathfrak{a}$  is a nonzero element in the kernel of the multiplication map  $\mathfrak{a} \otimes_A A/\mathfrak{a} \rightarrow A/\mathfrak{a}$ .

Note that  $a \otimes 1$  being nonzero is guaranteed by the fact that  $A$  is an integral domain. Looks like arguments above can be generalized to any integral domain.

Geometrically, this is an embedding of varieties  $\text{Spec } A/\mathfrak{a} \hookrightarrow \text{Spec } A$ .

Refer to the Stacks projects 04PQ for general discussions on when a quotient ring  $R/J$  is flat over  $R$ . Refer to 04PU, 04PV and 04PW for an algebraic geometry perspective.

**Example 18** (Localizations). Any localization  $S^{-1}A$  of a ring  $A$  is always a flat  $A$ -module, because  $S^{-1}A \otimes_A \mathfrak{a}$  is naturally isomorphic to  $S^{-1}\mathfrak{a}$ .

[TODO] Geometric examples.

**Example 19** (A non-flat projection). Let  $k$  be a field and  $A = k[x, y]/(xy)$ ,  $B = k[t]$ . Define an  $B$ -module structure on  $A$  via the homomorphism  $\varphi : B \rightarrow A$ ,  $t \mapsto x$  (that is, identifying  $x$  in  $A$  with  $t$ ). Then  $A$  is not a flat  $B$ -module.

This is because, if we pick  $\mathfrak{t} := (t) = tB$  an ideal of  $B$ , then  $y$  is a  $t$ -torsion element in  $A$  (by  $t \cdot y = \varphi(t)y = xy = 0$ ). That is,  $t \otimes_t y$  is a nonzero element in the kernel of the multiplication map  $\mathfrak{t} \otimes_B A \rightarrow A$ .

Geometrically, this corresponds to projecting the union of two coordinate axes  $\text{Spec } A$  onto the  $x$ -axis, i.e.  $\text{Spec } \varphi : (x, y) \mapsto x$ . Picking  $\mathfrak{t} = (t)$  corresponds to looking at the fibers around  $t = 0$ . Note that the fiber over the origin  $(0, 0)$  is the entire  $y$ -axis, while fibers over other points are singletons. This “jumping” of fiber dimension is reflected algebraically by the non-flatness of  $B$  as an  $A$ -module.

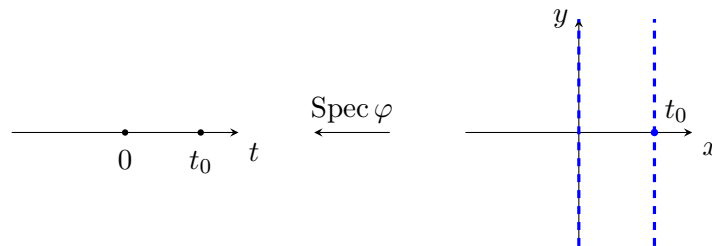


Figure 2:  $\text{Spec } \varphi$  and its fibers



**Example 20** (A non-flat parametrization). Let  $k$  be a field and  $A = k[x, y]/(y^2 - x^3)$ ,  $B = k[t]$ . Define an  $A$ -module structure on  $B$  via the homomorphism  $\varphi : A \rightarrow B$ ,  $(x, y) \mapsto (t^2, t^3)$ . Then  $B$  is not a flat  $A$ -module.

This is because, if we pick  $\mathfrak{a} := (x, y)$  an ideal of  $A$ , then  $x \cdot t = y \cdot 1_B$  becomes a nontrivial  $\mathfrak{a}$ -relation in  $B$ , i.e.  $x \otimes_A t - y \otimes_A 1_B$  is a nonzero element in the kernel of the multiplication map  $\mathfrak{a} \otimes_A B \rightarrow B$ .

Geometrically, this corresponds to the fact that the parametrized curve  $\text{Spec } \varphi : t \mapsto (t^2, t^3)$  has a cusp at the origin. Picking  $\mathfrak{a} = (x, y)$  corresponds to looking at the fibers of this map around the origin. [TODO] Understand this thoroughly.

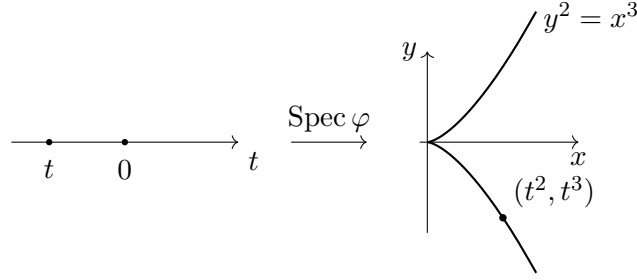


Figure 3:  $\text{Spec } \varphi$

**Exercise 21** ([AM69]-exr-2.27: Absolute flatness). A ring  $A$  is absolutely flat (or von Neumann regular) if every  $A$ -module is flat. Show that TFAE:

- i)  $A$  is absolutely flat;
- ii) every principal ideal of  $A$  is idempotent;
- iii) every finitely generated ideal of  $A$  is a direct summand of  $A$ .

*Proof.* i  $\implies$  ii: Consider the commutative diagram

$$\begin{array}{ccc} (x) & \longrightarrow & (x) \otimes A/(x) \\ \downarrow & & \uparrow \\ A & \longrightarrow & A/(x) \end{array}$$

As inclusion, the left vertical arrow is injective, hence so do the right vertical arrow. As projection, the bottom horizontal arrow is surjective, hence so do the top horizontal arrow. Note that  $(x) \rightarrow A \rightarrow A/(x)$  is zero, this forces the top horizontal arrow to be zero. But it is also surjective, hence  $(x) \otimes A/(x) = 0$ . Now note that  $(x) \otimes A/(x) \cong (x)/(x^2)$ , hence  $(x) = (x^2)$ .

ii  $\implies$  iii: First we show that every principal ideal  $(x)$  is generated by an idempotent element  $e$ . Since  $(x) = (x^2)$ , we have  $x = ax^2$ , then  $e := ax$  is idempotent and  $(x) = (e)$ . Next, for any two principal ideals  $(e)$  and  $(f)$  generated by idempotent elements  $e$  and  $f$ , we have  $(e, f) = (e + f - ef)$  which is also generated by an idempotent element. Then inductively we know that any finitely generated ideal is generated by an idempotent element, hence is a direct summand of  $A$  since  $A \cong A/(e) \oplus A/(1 - e) \cong (1 - e) \oplus (e)$ .

iii  $\implies$  i: By Exercise 16, to verify any  $A$ -module  $M$  is flat, it suffices to verify the flatness for any finitely generated ideal  $\mathfrak{a}$ . Say  $\mathfrak{a} \oplus \mathfrak{b} = A$ , then

$$0 \longrightarrow \mathfrak{a} \longrightarrow A$$

tensoring  $M$  gives the sequence

$$0 \longrightarrow \mathfrak{a} \otimes M \longrightarrow A \otimes M = \mathfrak{a} \otimes M \oplus \mathfrak{b} \otimes M$$

hence is exact. □

**Exercise 22** ([AM69]-exr-2.28). A Boolean ring is absolutely flat.

*Proof.* Because in a Boolean ring, every element is idempotent.  $\square$

## 2 Chapter 3

**Exercise 23** ([AM69]-exr-3.2). Let  $\mathfrak{a}$  be an ideal of a ring  $A$ , and let  $S = 1 + \mathfrak{a}$ . Show that  $S^{-1}\mathfrak{a}$  is contained in the Jacobson radical of  $S^{-1}A$ .

*Proof.* Easily verify that  $S$  is an multiplicative set. It suffices to prove for all  $\frac{a_1}{1+a_2} \in S^{-1}\mathfrak{a}$ ,  $1 + \frac{a_1}{1+a_2} \cdot \frac{x}{1+a_3}$  is a unit in  $S^{-1}A$  for all  $\frac{x}{1+a_3} \in S^{-1}A$ . The inverse  $\frac{(1+a_2)(1+a_3)}{(1+a_2)(1+a_3)+a_1x}$  would do.

Proving [AM69]-cor-2.5:  $M$  f.g.,  $\mathfrak{a}$  is an ideal of  $A$  s.t.  $\mathfrak{a}M = M$ . Taking  $S^{-1}$  gives

$$(S^{-1}\mathfrak{a})(S^{-1}M) = (S^{-1}\mathfrak{a})M = S^{-1}M$$

By Nakayama's lemma,  $S^{-1}M = 0$ . This shows that  $\exists s \in S$  s.t.  $sM = 0$  (by [AM69]-exr-3.1). Say  $s = 1 + a$ . Done.  $\square$

**Exercise 24** ([AM69]-exr-3.12: Torsion and torsion-free modules). Let  $A$  be an integral domain,  $M$  be a finitely generated  $A$ -module, and  $T(M)$  the torsion submodule of  $M$ . Show that

1.  $T(M)$  is a indeed a submodule of  $M$ .
2.  $M/T(M)$  is torsion-free.
3. Module homomorphism  $f : M \rightarrow N$  maps  $T(M)$  into  $T(N)$ .
4. Taking torsion is an left exact functor.
5. Let  $K$  be the field of fractions of  $A$ . Then  $T(M)$  is exactly the kernel of the localization map  $M \rightarrow K \otimes_A M$ . Equivalently, there is a natural isomorphism  $T(M) \cong \text{Tor}_1^A(K/A, M)$ .

*Proof.*

1. For any  $x, y \in T(M)$ ,  $\exists a, b \in A \setminus \{0\}$  s.t.  $ax = 0$ ,  $by = 0$ . Then  $ab(x+y) = abx + aby = 0 + 0 = 0$ . Thus  $x+y \in T(M)$ . Similarly for scalar multiplication.
2. For any  $x \in M$ , if  $\bar{x} \in M/T(M)$  is torsion, then  $\exists a \in A \setminus \{0\}$  s.t.  $a \cdot \bar{x} = \bar{0}$ . This means  $ax \in T(M)$ , hence  $x \in T(M)$  and  $\bar{x} = \bar{0}$ . Thus  $M/T(M)$  is torsion-free.
3. For any  $x \in T(M)$ ,  $\exists a \in A \setminus \{0\}$  s.t.  $ax = 0$ . Then  $af(x) = f(ax) = f(0) = 0$ . Thus  $f(x) \in T(N)$ .
4. Say  $0 \rightarrow M_1 \xrightarrow{f} M_2 \xrightarrow{g} M_3$  is exact. We need to show that  $0 \rightarrow T(M_1) \xrightarrow{f} T(M_2) \xrightarrow{g} T(M_3)$  is exact.

First,  $g \circ f = 0$  passes to the torsion submodules.

Secondly,  $f$  is injective on  $T(M_1)$  by the injectivity of  $f$  on  $M_1$ .

Finally, for any  $y \in T(M_2)$  with  $g(y) = 0$ , by the exactness of the original sequence, there exists  $x \in M_1$  s.t.  $f(x) = y$ . Since  $y$  is torsion,  $\exists b \in A \setminus \{0\}$  s.t.  $b \cdot y = 0$ . Then  $0 = b \cdot y = b \cdot f(x) = f(b \cdot x)$ , which implies  $b \cdot x = 0$  by the injectivity of  $f$ . Thus  $x \in T(M_1)$  and  $f(x) = y$ . This shows the exactness at  $T(M_2)$ .

5. See Section 1.1, where we systematically treat the etymology of  $\text{Tor}$ .

$\square$

**Exercise 25** ([AM69]-exr-3.13). Taking torsion commutes with localization, i.e. for any multiplicative set  $S$  of  $A$ ,  $T(S^{-1}M) \cong S^{-1}T(M)$ .

As a consequence, show that torsion-free is a local property.

*Proof.* Let  $k$  be the field of fractions of  $A$ . By Exercise 24, consider the exact sequence

$$0 \longrightarrow T(M) \longrightarrow M \longrightarrow k \otimes M$$

Taking localization  $S^{-1}$  gives the exact sequence

$$0 \longrightarrow S^{-1}T(M) \longrightarrow S^{-1}M \longrightarrow k \otimes M$$

Consider another exact sequence

$$0 \longrightarrow T(S^{-1}M) \longrightarrow S^{-1}M \longrightarrow k \otimes S^{-1}M \cong k \otimes M$$

Note that both  $S^{-1}T(M)$  and  $T(S^{-1}M)$  are realized as the kernels of the same map  $S^{-1}M \rightarrow k \otimes M$ . Thus they are naturally isomorphic.

For the consequence, say for all maximal ideals  $\mathfrak{m}$  of  $A$ ,  $M_{\mathfrak{m}}$  is torsion-free. Then by the isomorphism above,  $T(M)_{\mathfrak{m}} \cong T(M_{\mathfrak{m}}) = 0$ . Being a zero module is a local property, thus  $M$  is torsion-free.  $\square$

**Example 26** ([AM69]-exm-3.19: Support of a module). Let  $A$  be a ring,  $M$  an  $A$ -module. Define the support of  $M$  to be the set  $\text{Supp } M$  of prime ideals  $\mathfrak{p}$  of  $A$  such that  $M_{\mathfrak{p}} \neq 0$ .

Show that

1.  $M = 0$  iff  $\text{Supp } M = \emptyset$ .
2.  $V(\mathfrak{a}) = \text{Supp}(A/\mathfrak{a})$ .
3. If  $0 \rightarrow M_1 \rightarrow M_2 \rightarrow M_3 \rightarrow 0$  is exact, then  $\text{Supp } M_2 = \text{Supp } M_1 \cup \text{Supp } M_3$ .
4. If  $M = \sum_i M_i$ , then  $\text{Supp } M = \bigcup_i \text{Supp } M_i$ .
5. If  $M$  is finitely generated, then  $\text{Supp } M = V(\text{Ann}(M))$  (and hence is a closed subset of  $\text{Spec } A$ ).
6. If  $M$  and  $N$  are finitely generated, then  $\text{Supp}(M \otimes_A N) = \text{Supp } M \cap \text{Supp } N$ .
7. If  $M$  is finitely generated and  $\mathfrak{a}$  is an ideal of  $A$ , then  $\text{Supp}(M/\mathfrak{a}M) = V(\mathfrak{a} + \text{Ann}(M))$ .
8. If  $f : A \rightarrow B$  is a ring homomorphism and  $M$  is a finitely generated  $A$ -module, then  $\text{Supp}(B \otimes_A M) = (\text{Spec } f)^{-1}(\text{Supp } M)$ .

*Proof.*

1. If  $M = 0$ , then for all prime ideal  $\mathfrak{p}$ ,  $M_{\mathfrak{p}} = 0$ . Conversely, if  $\text{Supp } M = \emptyset$ , then for all prime ideal  $\mathfrak{p}$ ,  $M_{\mathfrak{p}} = 0$ . This implies that for all  $x \in M$ ,  $\exists s \in A \setminus \mathfrak{p}$  s.t.  $sx = 0$ . Thus the annihilator ideal  $\text{Ann}(x)$  is not contained in any prime ideal, which forces  $\text{Ann}(x) = A$  and hence  $x = 0$ . Thus  $M = 0$ .
2. For any prime ideal  $\mathfrak{p} \in V(\mathfrak{a})$ ,  $\mathfrak{a} \subseteq \mathfrak{p}$ . Thus the annihilated part of  $A_{\mathfrak{p}}$  never touches  $\mathfrak{a}$  and hence  $(A/\mathfrak{a})_{\mathfrak{p}} \neq 0$ . Conversely, for any prime ideal  $\mathfrak{p} \notin V(\mathfrak{a})$ ,  $\exists a \in \mathfrak{a} \setminus \mathfrak{p}$ . Then  $a/1$  is a unit in  $A_{\mathfrak{p}}$ , which forces  $(A/\mathfrak{a})_{\mathfrak{p}} = 0$ .
3. By the exactness of localization, we have the exact sequence

$$0 \longrightarrow (M_1)_{\mathfrak{p}} \longrightarrow (M_2)_{\mathfrak{p}} \longrightarrow (M_3)_{\mathfrak{p}} \longrightarrow 0$$

for all prime ideal  $\mathfrak{p}$ . Thus  $(M_2)_{\mathfrak{p}} \neq 0$  iff either  $(M_1)_{\mathfrak{p}} \neq 0$  or  $(M_3)_{\mathfrak{p}} \neq 0$ .

4. Localization, as a special tensor functor, commutes with taking sums (the later is a colimit, or more concretely, a direct limit) [TODO: Detail this nonsense] (A concrete proof can be found in Exercise 10). Thus for all prime ideal  $\mathfrak{p}$ , we have

$$M_{\mathfrak{p}} = \left( \sum_i M_i \right)_{\mathfrak{p}} = \sum_i (M_i)_{\mathfrak{p}}$$

Hence the conclusion follows.

5. Say  $M = \sum_i M_i$  where each  $M_i$  is cyclic, i.e.  $M_i \cong A/\text{Ann}(m_i)$  for some  $m_i \in M_i$ . Then by 4, we have

$$\text{Supp } M = \bigcup_i \text{Supp } M_i = \bigcup_i V(\text{Ann}(m_i)) = V\left(\bigcap_i \text{Ann}(m_i)\right) = V(\text{Ann}(M))$$

Note that the second last equality holds by the finiteness of the generating set  $\{m_i\}$ .

6. Recall that localization commutes with tensor product, i.e. for all prime ideal  $\mathfrak{p}$ ,

$$(M \otimes_A N)_{\mathfrak{p}} \cong M_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} N_{\mathfrak{p}}$$

Thus by Exercise 2,  $(M \otimes_A N)_{\mathfrak{p}} = 0$  iff both  $M_{\mathfrak{p}} = 0$  or  $N_{\mathfrak{p}} = 0$ . The conclusion follows.

7. We have

$$\begin{aligned} \text{Supp}(M/\mathfrak{a}M) &= \text{Supp}(A/\mathfrak{a} \otimes_A M) = \text{Supp}(A/\mathfrak{a}) \cap \text{Supp}(M) \\ &= V(\mathfrak{a}) \cap V(\text{Ann}(M)) = V(\mathfrak{a} + \text{Ann}(M)) \end{aligned}$$

8. Say  $\mathfrak{q} \in \text{Spec } B$ , and denote  $\mathfrak{p} := f^{-1}(\mathfrak{q})$ , we are to show that  $(B \otimes_A M)_{\mathfrak{q}} = 0$  iff  $M_{\mathfrak{p}} = 0$ . It's clear that if  $M_{\mathfrak{p}} = 0$ , then  $(B \otimes_A M)_{\mathfrak{q}} = 0$  by the isomorphism

$$(B \otimes_A M)_{\mathfrak{q}} \cong B_{\mathfrak{q}} \otimes_{A_{\mathfrak{p}}} M_{\mathfrak{p}}$$

Conversely, if  $(B \otimes_A M)_{\mathfrak{q}} = 0$ , by the isomorphism above we have  $B_{\mathfrak{q}} \otimes_{A_{\mathfrak{p}}} M_{\mathfrak{p}} = 0$ . Note that we can't use Exercise 2 since  $B_{\mathfrak{p}}$  may not be finitely generated over  $A_{\mathfrak{p}}$ . However, tensoring with the residue field  $A_{\mathfrak{p}}/\mathfrak{p}$ , we still have  $B_{\mathfrak{q}}/\mathfrak{q}B_{\mathfrak{q}} \otimes_{A_{\mathfrak{p}}} (M_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}}) = 0$ , where  $M_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}}$  is a finite-dimensional vector space over the field  $A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}}$ , and the whole LHS is a finite-dimensional vector space over the field  $B_{\mathfrak{q}}/\mathfrak{q}B_{\mathfrak{q}}$ . Thus it must be of zero dimension. In turn  $M_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}} = 0$ . By Nakayama's lemma,  $M_{\mathfrak{p}} = 0$ .

□

*Remark.* [TODO] Best appreciated in a geometric view. If  $B$  is an  $A$ -algebra defined by the ring map  $\varphi : A \rightarrow B$ , then  $\text{Supp}_A B$  is exactly the image of  $\text{Spec } \varphi : \text{Spec } B \rightarrow \text{Spec } A$ , since for any prime ideal  $\mathfrak{p}$  of  $A$ , the fiber over  $\mathfrak{p}$  is  $\text{Spec}(B \otimes_A k(\mathfrak{p}))$ , which is nonempty iff  $B \otimes_A k(\mathfrak{p}) \neq 0$  iff  $B_{\mathfrak{p}} \neq 0$ .

**Exercise 27** ([AM69]-exr-3.20).  $f : A \rightarrow B$  ring homomorphism. Show that

1. Every prime ideal of  $A$  is a contracted ideal iff  $\text{Spec } f$  is surjective.
2. Every prime ideal of  $B$  is an extended ideal implies that  $\text{Spec } f$  is injective.
3. The converse of (2) is not true in general.

*Proof.*

1. By definition,  $\mathfrak{p} \in \operatorname{Spec} A$  is a contracted ideal iff  $\mathfrak{p} = f^{-1}(\mathfrak{q})$  for some prime ideal  $\mathfrak{q}$  of  $B$ , exactly the surjectivity of  $\operatorname{Spec} f$ .
2. Denote  $\mathfrak{a}^e$  for ideal extension of  $\mathfrak{a}$  alongside  $f$ , and  $\mathfrak{b}^c$  for contraction. For any  $\mathfrak{q} \in \operatorname{Spec} B$ , we have  $\mathfrak{q} = \mathfrak{a}^e$  for some ideal  $\mathfrak{a}$  of  $A$ . This makes  $\mathfrak{q} \mapsto \mathfrak{q}^{ce}$  an identity map on  $\operatorname{Spec} B$  by the fact that  $\mathfrak{q}^{ce} = \mathfrak{a}^{ece} = \mathfrak{a}^e = \mathfrak{q}$ . Hence  $\operatorname{Spec} f : \mathfrak{q} \mapsto \mathfrak{q}^c$  is injective.
3. Consider  $f : \mathbb{Z} \rightarrow \mathbb{Q}$ .  $\operatorname{Spec} f$  is injective since  $\mathbb{Q}$  is a field. However, the prime ideal  $(0)$  of  $\mathbb{Q}$  cannot be extended from any ideal of  $\mathbb{Z}$ .

□

*Remark.* If  $f$  is an integral extension, then the converse of (2) holds true. (By the going-up / lying-over / incomparability theorems.)

**Exercise 28** ([AM69]-exr-3.21).

1. Let  $A$  be a ring,  $S$  a multiplicative set of  $A$ ,  $\varphi : A \rightarrow S^{-1}A$  the localization homomorphism. Show that the map  $\operatorname{Spec} \varphi : \operatorname{Spec} S^{-1}A \rightarrow \operatorname{Spec} A$  is a homeomorphism onto its image in  $X = \operatorname{Spec} A$ . Denote this image by  $S^{-1}X$ .  
In particular, if  $f \in A$ , then the map  $\operatorname{Spec} A_f \rightarrow \operatorname{Spec} A$  is a homeomorphism onto the basic open subset  $D_A(f)$  of  $X$ .
2. Let  $f : A \rightarrow B$  be a ring homomorphism. Let  $X = \operatorname{Spec} A$  and  $Y = \operatorname{Spec} B$ ,  $\operatorname{Spec} f : \operatorname{Spec} B \rightarrow \operatorname{Spec} A$ . Identify  $S^{-1}X \subseteq X$  and  $S^{-1}Y \subseteq Y$ . Show that  $\operatorname{Spec}(S^{-1}f) : \operatorname{Spec} S^{-1}B \rightarrow \operatorname{Spec} S^{-1}A$  is the restriction of  $\operatorname{Spec} f$  to  $S^{-1}Y$ , and that  $S^{-1}Y = (\operatorname{Spec} f)^{-1}(S^{-1}X)$ .
3. Let  $\mathfrak{a}$  be an ideal of  $A$  and  $\mathfrak{b} = \mathfrak{a}^e$  its extension alongside  $f : A \rightarrow B$ . Let  $\bar{f} : A/\mathfrak{a} \rightarrow B/\mathfrak{b}$  be the induced homomorphism. Identify  $\operatorname{Spec}(A/\mathfrak{a}) = V(\mathfrak{a}) \subseteq X$  and  $\operatorname{Spec}(B/\mathfrak{b}) = V(\mathfrak{b}) \subseteq Y$ , show that  $\operatorname{Spec}(\bar{f})$  is the restriction of  $\operatorname{Spec} f$  to  $V(\mathfrak{b})$ .
4. Let  $\mathfrak{p} \in \operatorname{Spec} A$ . Take  $S = A \setminus \mathfrak{p}$  in (2) and reduce mod  $S^{-1}\mathfrak{p}$  as in (3). Deduce that the subspace  $(\operatorname{Spec} f)^{-1}(\mathfrak{p})$  of  $\operatorname{Spec} B$  is homeomorphic to  $\operatorname{Spec}(B \otimes_A k(\mathfrak{p}))$ , the fiber of  $\operatorname{Spec} f$  over  $\mathfrak{p}$ .

*Proof.*

1. Continuity and onto-ness are clear. Injectivity follows from Exercise 28 and the fact that every prime ideal of  $S^{-1}A$  is an extended ideal from  $A$ . It suffices to show that  $\operatorname{Spec} \varphi$  is an open map onto its image.

Fix some  $f/s \in S^{-1}A$ . Consider any contracted ideal  $\mathfrak{p} \in S^{-1}X$ . Then it can be contracted from  $S^{-1}\mathfrak{p}$ . Suppose  $S^{-1}\mathfrak{p} \in D_{S^{-1}A}(f/s)$ , i.e.  $f/s \notin S^{-1}\mathfrak{p}$  or equivalently any  $t \in S$  won't make  $f \in (\mathfrak{p} : t)$ . Note that  $\mathfrak{p}$  is contracted, hence  $(\mathfrak{p} : t) = \mathfrak{p}$ . Thus  $f \notin \mathfrak{p}$ , i.e.  $\mathfrak{p} \in D_A(f)$ . Above arguments can be reversed, hence we have shown that

$$\operatorname{Spec} \varphi(D_{S^{-1}A}(f/s)) = D_A(f) \cap S^{-1}X$$

2. Apply the  $\operatorname{Spec}$  functor to the commutative diagram gives Figure 4.

The equation  $S^{-1}Y = (\operatorname{Spec} f)^{-1}(S^{-1}X)$  is given by the fact that

$$\begin{aligned}
& \mathfrak{q} \in S^{-1}Y \\
& \iff f(S) \cap \mathfrak{q} = \emptyset \\
& \iff \forall s \in S, f(s) \notin \mathfrak{q} \\
& \iff \forall s \in S, s \notin f^{-1}(\mathfrak{q}) \\
& \iff S \cap f^{-1}(\mathfrak{q}) = \emptyset \\
& \iff f^{-1}(\mathfrak{q}) \in S^{-1}X \\
& \iff \mathfrak{q} \in (\operatorname{Spec} f)^{-1}(S^{-1}X)
\end{aligned}$$

$$\begin{array}{ccc}
A & \xrightarrow{f} & B \\
\downarrow \varphi & & \downarrow \psi \\
S^{-1}A & \xrightarrow{S^{-1}f} & (f(S))^{-1}B
\end{array}
\qquad
\begin{array}{ccc}
X & \xleftarrow{\operatorname{Spec} f} & Y \\
\subseteq \uparrow & & \subseteq \uparrow \\
S^{-1}X & \xleftarrow{\operatorname{Spec}(S^{-1}f)} & S^{-1}Y
\end{array}$$

Figure 4

3. Similar to (2), see Figure 5.

$$\begin{array}{ccc}
A & \xrightarrow{f} & B \\
\downarrow \varphi & & \downarrow \psi \\
A/\mathfrak{p} & \xrightarrow{\bar{f}} & A/\mathfrak{p} \otimes_A B
\end{array}
\qquad
\begin{array}{ccc}
X & \xleftarrow{\operatorname{Spec} f} & Y \\
\subseteq \uparrow & & \subseteq \uparrow \\
V_A(\mathfrak{p}) & \xleftarrow{\operatorname{Spec}(\bar{f})} & V_B(\mathfrak{p}^e)
\end{array}$$

Figure 5

4. Analyzing Figure 6 gives the desired homeomorphism.

$$\begin{array}{ccc}
A & \xrightarrow{f} & B \\
\downarrow & & \downarrow \\
A_{\mathfrak{p}} & \xrightarrow{f_{\mathfrak{p}}} & B_{\mathfrak{p}} \\
\downarrow & & \downarrow \\
k(\mathfrak{p}) & \longrightarrow & k(\mathfrak{p}) \otimes_A B
\end{array}
\qquad
\begin{array}{ccc}
\operatorname{Spec} A & \xleftarrow{\operatorname{Spec} f} & \operatorname{Spec} B \\
\subseteq \uparrow & & \subseteq \uparrow \\
\operatorname{Spec} A_{\mathfrak{p}} & \xleftarrow{\operatorname{Spec}(f_{\mathfrak{p}})} & \operatorname{Spec} B_{\mathfrak{p}} \\
\subseteq \uparrow & & \subseteq \uparrow \\
\{(0)\} & \longleftarrow & \operatorname{Spec}(k(\mathfrak{p}) \otimes_A B)
\end{array}$$

Figure 6

□

### 3 Chapter 4

**Exercise 29** ([AM69]-exr-4.4). In the polynomial ring  $\mathbb{Z}[t]$ , the ideal  $\mathfrak{m} = (2, t)$  is maximal and the ideal  $\mathfrak{q} = (4, t)$  is  $\mathfrak{m}$ -primary, but is not a power of  $\mathfrak{m}$ .

*Proof.*  $\mathfrak{m}$  being maximal is clear.  $\sqrt{\mathfrak{q}} = (2, t)$  is also clear. For any  $ab \in \mathfrak{q}$ , if  $a \notin \mathfrak{m}$ , then  $a$  is an odd integer. Hence  $4 \mid b$  or  $t \mid b$ , i.e.  $b \in \mathfrak{q}$ , showing that  $\mathfrak{q}$  is  $\mathfrak{m}$ -primary. Finally, note that  $(2, t)^2 = (4, 2t, t^2) \neq (4, t) = \mathfrak{q}$ . □

**Exercise 30** ([AM69]-exr-4.5). Consider the polynomial ring  $K[x, y, z]$  where  $K$  is a field. Let  $\mathfrak{p}_1 = (x, y)$ ,  $\mathfrak{p}_2 = (x, z)$ ,  $\mathfrak{m} = (x, y, z)$ . Note that  $\mathfrak{p}_1$  and  $\mathfrak{p}_2$  are prime ideals, and  $\mathfrak{m}$  is a maximal ideal. Let  $\mathfrak{a} = \mathfrak{p}_1\mathfrak{p}_2 = (x^2, xy, xz, yz)$ . Show that  $\mathfrak{a} := \mathfrak{p}_1 \cap \mathfrak{p}_2 \cap \mathfrak{m}^2$  is a reduced primary decomposition of  $\mathfrak{a}$ . Identify the associated primes of  $\mathfrak{a}$ .

*Proof.* The mutual incomparability of  $\mathfrak{p}_1$ ,  $\mathfrak{p}_2$  and  $\mathfrak{m}$  is clear.  $\mathfrak{p}_1$  and  $\mathfrak{p}_2$  are minimal and  $\mathfrak{m}$  is embedded.  $\square$

**Exercise 31** ([AM69]-exr-4.6). Let  $X$  be an infinite compact Hausdorff topological space,  $C(X)$  be the ring of real-valued continuous functions on  $X$  (cf. [AM69]-exr-1.26). Is the zero ideal decomposable in this ring?

*Proof.* It can't be. If it is, recall that by the remark below [AM69, proposition 4.7], the set of zero-divisors of  $C(X)$  is exactly the union of the associated primes. Consider the maximal ideals in these associated primes, they correspond to the points in  $X$  by [AM69, exr-1.26], say  $x_1, x_2, \dots, x_n$ . Then every zero-divisor  $f$  must vanish at one of these points, i.e.  $f(x_i) = 0$  for some  $1 \leq i \leq n$ . However, since  $X$  is infinite, we can always find some point  $x \in X \setminus \{x_1, x_2, \dots, x_n\}$ . To reach a contradiction, we shall construct a continuous function  $g$  on  $X$  such that  $g(x_i) \neq 0$  for all  $1 \leq i \leq n$ , but still a zero-divisor.

By the Hausdorff property, for each  $1 \leq i \leq n$ , we can find some open neighborhood  $U_i$  of  $x_i$  and some open neighborhood  $V_i$  of  $x$  such that  $U_i \cap V_i = \emptyset$ . Let  $U := \bigcup_{i=1}^n U_i$ ,  $V := \bigcap_{i=1}^n V_i$ , then  $U$  and  $V$  are still open neighborhoods of  $\{x_1, x_2, \dots, x_n\}$  and  $x$  respectively, and  $U \cap V = \emptyset$ . By the Urysohn lemma, there exists some continuous function  $g : X \rightarrow [0, 1]$  such that  $g(x_i) = 1$  for all  $1 \leq i \leq n$  and  $g|_{X \setminus U} = 0$ . It suffices to show that  $g$  is a zero-divisor. It's counterpart can be constructed in a symmetric way: By Urysohn lemma again, there exists a continuous function  $h : X \rightarrow [0, 1]$  such that  $h(x) = 1$  and  $h|_{X \setminus V} = 0$ . Then  $gh = 0$ , showing that  $g$  is a zero-divisor.  $\square$

**Exercise 32** ([AM69]-exr-4.7).  $A$  is a ring.

1.  $\mathfrak{a}[x]$  is the extension of  $\mathfrak{a}$  alongside the inclusion  $A \rightarrow A[x]$ .
2. If  $\mathfrak{p}$  is a prime ideal in  $A$ , then  $\mathfrak{p}[x]$  is a prime ideal in  $A[x]$ .
3. If  $\mathfrak{q}$  is  $\mathfrak{p}$ -primary in  $A$ , then  $\mathfrak{q}[x]$  is  $\mathfrak{p}[x]$ -primary in  $A[x]$ .
4. If  $\mathfrak{a} = \bigcap_{i=1}^n \mathfrak{q}_i$  is a primary decomposition of  $\mathfrak{a}$  in  $A$ , then  $\mathfrak{a}[x] = \bigcap_{i=1}^n \mathfrak{q}_i[x]$  is a primary decomposition of  $\mathfrak{a}[x]$  in  $A[x]$ .
5. If  $\mathfrak{p}$  is a minimal prime ideal of  $\mathfrak{a}$ , then  $\mathfrak{p}[x]$  is a minimal prime ideal of  $\mathfrak{a}[x]$ .

*Proof.*

1. Trivial.
2.  $A[x]/\mathfrak{p}[x] \cong (A/\mathfrak{p})[x]$  is an integral domain since  $A/\mathfrak{p}$  is so.
3.  $A[x]/\mathfrak{q}[x] \cong (A/\mathfrak{q})[x]$ . Any zero-divisor in  $(A/\mathfrak{q})[x]$  must have all its coefficients being zero-divisors in  $A/\mathfrak{q}$  by [AM69]-exr-1.2-iii, hence nilpotent since  $\mathfrak{q}$  is  $\mathfrak{p}$ -primary. Thus this zero-divisor is nilpotent by [AM69]-exr-1.2-ii, and  $\mathfrak{q}[x]$  is  $\mathfrak{p}[x]$ -primary. It's radical is at least  $\mathfrak{p}[x]$  because  $\mathfrak{q}$  can reach  $\mathfrak{p}$  by taking radicals. Regarding that  $\mathfrak{p}[x]$  is radical, we have  $r(\mathfrak{q}[x]) = \mathfrak{p}[x]$ .
4. There is nothing to prove.
5. There is nothing to prove.

$\square$

**Exercise 33** ([AM69]-exr-4.20: Primary decomposition of modules). Let  $A$  be a ring,  $M$  be a fixed  $A$ -module,  $N$  a submodule of  $M$ . The radical of  $N$  in  $M$  is defined to be

$$r_M(N) = \{x \in A : x^q M \subseteq N \text{ for some } q > 0\}$$

Show that  $r_M(N) = r(N : M) = r(\text{Ann}(M/N))$ .

*Proof.* Recall that  $(N : M) = \{a \in A : aM \subseteq N\}$ , hence the first equality is straightforward. The second equality follows from the fact that  $\text{Ann}(M/N) = (N : M)$ .  $\square$

*Remark.* Analogous to [AM69]-exr-1.13, we have

- $r_M(N) \supseteq (N : M)$
- $r(r_M(N)) = r_M(N)$
- $r_M(N \cap P) = r_M(N) \cap r_M(P)$
- $r_M(N) = A$  iff  $M = N$
- $r_M(N + P) \supseteq r(r_M(N) + r_M(P))$

Note that the last identity may not be an equality in general. For example, let  $M = A \oplus A$ ,  $N = A \oplus 0$ ,  $P = 0 \oplus A$ . Then  $r_M(N) = r_M(P) = 0$ , while  $r_M(N + P) = r_M(M) = A$ .

**Exercise 34** ([AM69]-exr-4.21). For any  $x \in A$ ,  $\varphi_x$  denotes the multiplication endomorphism  $M \rightarrow M$ ,  $m \mapsto xm$ . Then  $x$  is said to be a zero-divisor on  $M$  iff  $\varphi_x$  is not injective, is a nilpotent on  $M$  iff  $\varphi_x$  is nilpotent. A submodule  $Q$  of  $M$  is primary in  $M$  if  $Q \neq M$  and every zero-divisor in  $M/Q$  is nilpotent.

Show that if  $Q$  is primary in  $M$ , then  $(Q : M)$  is a primary ideal and hence  $r_M(Q)$  is a prime ideal  $\mathfrak{p}$ . We say that  $Q$  is  $\mathfrak{p}$ -primary in  $M$ .

*Proof.* Recall that  $(Q : M) = \text{Ann}(M/Q)$ . Note that  $Q \neq M$  by definition of primary submodule. Say  $ab \in (Q : M)$  for some  $a, b \in A$ .

In case that  $a$  is a zero-divisor on  $M/Q$ , then by the primary-ness of  $Q$ ,  $a$  is nilpotent on  $M/Q$  and hence  $a^n \in (Q : M)$  for some  $n > 0$ . If  $a$  is strong enough to kill  $M/Q$ , then  $a \in (Q : M)$  and we are done. Otherwise  $b$  kills  $aM/Q \neq 0$ . Thus  $b$  is a zero-divisor on  $M/Q$ , hence nilpotent on  $M/Q$ , i.e.  $b \in r(Q : M)$ .

In case that  $a$  is not a zero-divisor on  $M/Q$ , then  $aM/Q \neq 0$ . Since  $ab$  kills  $M/Q$ ,  $b$  must kill  $aM/Q \neq 0$ . Thus  $b$  is a zero-divisor on  $M/Q$ , hence nilpotent on  $M/Q$ , i.e.  $b \in r(Q : M)$ .

This shows that  $(Q : M)$  is a primary ideal.  $\square$

**Exercise 35** ([AM69]-exr-4.22). A primary decomposition of  $N$  in  $M$  is a representation of  $N$  as an intersection

$$N = Q_1 \cap Q_2 \cap \cdots \cap Q_n$$

of primary submodules of  $M$ ; it is a minimal primary decomposition if the ideals  $\mathfrak{p}_i = r_M(Q_i)$  are all distinct and if none of the components  $Q_i$  can be omitted from the intersection.

Prove that the prime ideals  $\mathfrak{p}_i$  depend only on  $N$  (and  $M$ ), not on the particular minimal primary decomposition of  $N$  in  $M$ .

*Proof.* Note that  $(N : M) = \bigcap_{i=1}^n (Q_i : M)$ . By Exercise 34, each  $(Q_i : M)$  is a  $\mathfrak{p}_i$ -primary ideal. Thus by the uniqueness of primary decomposition of ideals, the set  $\{\mathfrak{p}_i\}$  is uniquely determined by  $(N : M)$  and hence by  $N$  and  $M$ .  $\square$

## 4 Chapter 5

**Exercise 36** ([AM69]-exr-5.1). Let  $f : A \rightarrow B$  be an integral homomorphism of rings. Show that  $\text{Spec } f : \text{Spec } B \rightarrow \text{Spec } A$  is a closed map, i.e. the image of a closed set is closed.

*Proof.* Let  $V(\mathfrak{b}) \subseteq \text{Spec } B$  be a closed set for some ideal  $\mathfrak{b}$  of  $B$ . We are to show that  $f^{-1}(V(\mathfrak{b}))$  is closed in  $\text{Spec } A$ . Note that

$$f^{-1}(V(\mathfrak{b})) = \{\mathfrak{p} \in \text{Spec } A : \exists \mathfrak{q} \in V(\mathfrak{b}), f^{-1}(\mathfrak{q}) = \mathfrak{p}\}$$



while

$$V(f^{-1}(\mathfrak{b})) = \{\mathfrak{p} \in \operatorname{Spec} A : f^{-1}(\mathfrak{b}) \subseteq \mathfrak{p}\}$$

Thus it suffices to show the two sets above are equal.

Write  $f : A \rightarrow f(A) \subseteq B$ , the former is surjective and the latter is an integral extension. We have the ideal one-to-one correspondence between all ideals containing  $\ker f$  of  $A$ , and all ideals of  $f(A)$ . By [AM69, theorem 5.10], every prime ideal of  $f(A)$  can be contracted from some prime ideal of  $B$ . To summarize, every prime ideal of  $A$  containing  $\ker f$  can be contracted from some prime ideal of  $B$ . We call this the lying-over theorem.

“ $\subseteq$ ”: For any  $\mathfrak{p} \in f^{-1}(V(\mathfrak{b}))$ , there exists some  $\mathfrak{q} \in V(\mathfrak{b})$  such that  $f^{-1}(\mathfrak{q}) = \mathfrak{p}$ . Thus  $f^{-1}(\mathfrak{b}) \subseteq f^{-1}(\mathfrak{q}) = \mathfrak{p}$ , i.e.  $\mathfrak{p} \in V(f^{-1}(\mathfrak{b}))$ .

“ $\supseteq$ ”: For any  $\mathfrak{p} \in V(f^{-1}(\mathfrak{b}))$ , we have  $\ker f \subseteq f^{-1}(\mathfrak{b}) \subseteq \mathfrak{p}$ . By the lying-over theorem, there exists some prime ideal  $\mathfrak{q}$  of  $B$  such that  $f^{-1}(\mathfrak{q}) = \mathfrak{p}$ . Note that  $\mathfrak{b} \subseteq \mathfrak{q}$  since  $f^{-1}(\mathfrak{b}) \subseteq \mathfrak{p} = f^{-1}(\mathfrak{q})$ . Thus  $\mathfrak{q} \in V(\mathfrak{b})$  and hence  $\mathfrak{p} \in f^{-1}(V(\mathfrak{b}))$ .  $\square$

**Exercise 37** ([AM69]-exr-5.2). Let  $A \subseteq B$  is an integral extension of rings. Let  $f : A \rightarrow \Omega$  be a homomorphism of  $A$  into an algebraically closed field  $\Omega$ . Show that there exists an extension of  $f$  to a homomorphism  $g : B \rightarrow \Omega$ .

*Proof.* By Zorn’s lemma it suffices to extend  $f$  to any intermediate ring  $A[x]$  where  $x \in B$ . Note that  $x$  is integral over  $A$ , hence there exists some monic polynomial  $p(t) \in A[t]$  such that  $p(x) = 0$ . Applying  $f$  to the coefficients of  $p(t)$  gives a polynomial  $f(p)(t) \in \Omega[t]$ . Since  $\Omega$  is algebraically closed, there exists some  $a \in \Omega$  such that  $f(p)(a) = 0$ . We extend  $f$  to  $g : A[x] \rightarrow \Omega$  by sending  $x \mapsto a$ . It’s straightforward to verify that  $g$  is a well-defined ring homomorphism.  $\square$

**Exercise 38** ([AM69]-exr-5.3). The  $A$ -algebra functor  $- \otimes_A C$  preserves integrality of  $A$ -algebra homomorphisms, i.e. if  $f : B_1 \rightarrow B_2$  is an integral homomorphism of  $A$ -algebras, then so is  $f \otimes \operatorname{id}_C : B_1 \otimes_A C \rightarrow B_2 \otimes_A C$  for any  $A$ -algebra  $C$ .

*Proof.* We are to show that any finite sum  $\sum_i b_i \otimes c_i \in B_2 \otimes_A C$  for some  $b_i \in B_2$  and  $c_i \in C$  is integral over  $B_1 \otimes_A C$ . Since the integral closure is closed under addition, it suffices to show any  $b \otimes c \in B_2 \otimes_A C$  is integral over  $B_1 \otimes_A C$ . By the integrality of  $f$ , there exists some monic polynomial  $t^n + \sum_{i=0}^{n-1} a_i t^i \in B_1[t]$  such that  $p(b) = 0$ . Now

$$(b \otimes c)^n + \sum_{i=0}^{n-1} (a_i \otimes 1)(b \otimes c)^i = \left( b^n + \sum_{i=0}^{n-1} a_i b^i \right) \otimes c^n = 0$$

$\square$

**Exercise 39** ([AM69]-exr-5.6). Let  $B_1, \dots, B_n$  be integral  $A$ -algebras. Show that  $\prod_{i=1}^n B_i$  is an integral  $A$ -algebra.

*Proof.* By that integral closure is closed under finite sums, it suffices to show that any element of the form  $(0, \dots, 0, b, 0, \dots, 0) \in \prod_{i=1}^n B_i$  is integral over  $A$ . Say  $b$  vanishes in some monic polynomial  $f \in B_i[t]$ . Then  $(0, \dots, 0, b, 0, \dots, 0)$  vanishes in the same polynomial, where each coefficient is embedded via the canonical homomorphism  $A \rightarrow \prod_{i=1}^n B_i$ .  $\square$

## 5 Chapter 7

**Exercise 40.** TFAE for a Noetherian local ring  $(A, \mathfrak{m}, k)$  and an  $A$ -module  $M$ :

1.  $M$  is free.
2.  $M$  is flat.

3.  $\mathfrak{m} \otimes M \rightarrow A \otimes M$  is injective.
4.  $\mathrm{Tor}_1^A(k, M) = 0$ .

*Proof.*

- (1)  $\implies$  (2) is by that free modules are projective (by pulling back basis) and projective modules are flat (by the currifcation property of tensor functor).
- (2)  $\iff$  (3) is by the definition of flatness.
- (3)  $\implies$  (4) is by the long exact sequence of Tor alongside the short exact sequence

$$0 \longrightarrow \mathfrak{m} \longrightarrow A \longrightarrow k \longrightarrow 0$$

It remains to show that (4)  $\implies$  (1). Cosider the exact sequence

$$0 \longrightarrow Q \longrightarrow A^m \longrightarrow M \longrightarrow 0$$

where  $m$  is the minimal number of generators of  $M$ . By Nakayama's lemma, such number is also the dimension of  $k \otimes M$  as a  $k$ -vector space. Applying  $k \otimes -$  gives the long exact sequence

$$\mathrm{Tor}_1^A(k, M) = 0 \longrightarrow k \otimes Q \longrightarrow k^m \longrightarrow k^m \longrightarrow 0$$

Hence  $k \otimes Q = 0$ . By Nakayama's lemma again, we have  $Q = 0$  and  $M \cong A^m$  is free.  $\square$

## 6 Chapter 9

**Exercise 41** ([AM69]-exr-9.9: Chinese Remainder Theorem over Dedekind domains). Let  $\mathfrak{a}_1, \dots, \mathfrak{a}_n$  be ideals and let  $x_1, \dots, x_n$  be elements in a Dedekind domain  $A$ . Then the system of congruences  $x \equiv x_i \pmod{\mathfrak{a}_i}$  for  $i = 1, \dots, n$  has a solution  $x \in A$  iff for all  $i, j$ ,  $x_i \equiv x_j \pmod{\mathfrak{a}_i + \mathfrak{a}_j}$ .

*Proof.* Note that this is equivalent of the exactness of the following sequence:

$$A \longrightarrow \bigoplus_{i=1}^n A/\mathfrak{a}_i \longrightarrow \bigoplus_{i < j} A/(\mathfrak{a}_i + \mathfrak{a}_j)$$

where the map on the left is the natural projection, and the map on the right sends  $(\bar{x}_1, \dots, \bar{x}_n)$  to  $(\bar{x}_i - \bar{x}_j)_{i < j}$ . The exactness is a local property, hence it suffices to assume  $A$  is a DVR since Dedekind domains are locally DVRs. In this case, each ideal  $\mathfrak{a}_i$  is of the form  $(m^{k_i})$  for some  $k_i \geq 0$ , where  $(m)$  is the unique maximal ideal of  $A$ . WLOG we assume  $k_1 \leq k_2 \leq \dots \leq k_n$ . Then for any  $i < j$ , we have  $\mathfrak{a}_i + \mathfrak{a}_j = (m^{k_i})$ . Thus the exactness follows from the exactness of

$$A \longrightarrow \bigoplus_{i=1}^n A/(m^{k_i}) \longrightarrow \bigoplus_{i < j} A/(m^{k_i})$$

Say  $(\bar{x}_1, \dots, \bar{x}_n)$  lies in the kernel of the right map. Then for any  $i < j$ ,  $m^{k_i} \mid x_i - x_j$ . Let  $\delta_{i,j}$  be its divisor. We construct the preimage  $x$  as follows:

$$x = x_1 + \delta_{1,2}m^{k_1} + \delta_{2,3}m^{k_2} + \dots + \delta_{n-1,n}m^{k_n}$$

It's strightforward to verify that it satisfy the congurence relation  $x \equiv x_i \pmod{m^{k_i}}$ .  $\square$

## 7 Chapter 11

**Exercise 42** ([AM69]-exr-11.1). Let  $f \in k[x_1, \dots, x_r]$  be an irreducible polynomial over an algebraically closed field  $k$ . A point  $P$  on the variety  $f(x) = 0$  is non-singular iff not all the partial derivatives  $\frac{\partial f}{\partial x_i}$  vanish at  $P$ . Let  $A = k[x_1, \dots, x_r]/(f)$ , and let  $\mathfrak{m}$  be the maximal ideal of  $A$  corresponding to  $P$ . Show that  $P$  is non-singular iff the localization  $A_{\mathfrak{m}}$  is a regular local ring.

*Proof.* WLOG we assume  $P$  is the origin. Write

$$f(x_1, \dots, x_r) = \sum_{i=1}^r \frac{\partial f}{\partial x_i} \Big|_P x_i + \text{higher order terms}$$

Note that via pulling back from  $A$  to  $k[x_1, \dots, x_r]$ , we have

$$\mathfrak{m}/\mathfrak{m}^2 \cong ((x_1, \dots, x_r) + (f)) / ((x_1, \dots, x_r)^2 + (f)) = (x_1, \dots, x_r) / \left( (x_1, \dots, x_r)^2 + \left( \sum_{i=1}^r \frac{\partial f}{\partial x_i} \Big|_P x_i \right) \right)$$

In case that all partial derivatives vanish at  $P$ , then the dimension of the RHS is  $r$  since  $x_1, \dots, x_r$  form a basis. Otherwise, a nontrivial linear combination of  $x_1, \dots, x_r$  lies in the denominator, hence the dimension of the RHS is  $r - 1$ .  $\square$

**Exercise 43.** Reformulate [AM69, theorem 11.1] in terms of the Grothendieck group  $K(A_0)$  (cf. [AM69]-exr-7.25)

*Solution.* Given a Noetherian graded ring  $A = \bigoplus_{n=0}^{+\infty} A_n$ , an additive function  $\lambda$  over all finitely-generated  $A_0$ -modules. By the universal property of Grothendieck group, for any additive group  $G$ ,  $\lambda$  induces a group homomorphism  $\tilde{\lambda} : K(A_0) \rightarrow G$ . Then

$$P(M, t) = \sum_{n=0}^{+\infty} \lambda(M_n) t^n = \sum_{n=0}^{+\infty} \tilde{\lambda}([M_n]) t^n$$

is a rational function of the form  $\frac{f(t)}{\prod_{i=1}^s (1-t^{k_i})}$ , where  $f(t) \in \mathbb{Z}[t]$  and  $k_i$  are positive integers.

[TODO] What's the significance of this reformulation?

**Exercise 44** ([AM69]-exr-11.6). Let  $A$  be a ring, not necessarily Noetherian. Show that

$$1 + \dim A \leq \dim A[x] \leq 1 + 2 \dim A$$

*Proof.* Note that for any maximal ideal  $\mathfrak{m}$  of  $A$ ,  $\mathfrak{m}[x] + (x)$  is a maximal ideal of  $A[x]$  because it is the kernel of  $A[x] \rightarrow A \rightarrow A/\mathfrak{m}$ . Now say we have an maximal chain of prime ideals in  $A$  of length  $\dim A$ . The top prime ideal must be an maximal ideal of  $A$ , say  $\mathfrak{m}$ . Lifting it to  $A[x]$ , we have a chain of prime ideals in  $A[x]$  of length  $\dim A$ , with top prime ideal  $\mathfrak{m}[x]$ . Extend this chain by adding one more prime ideal  $\mathfrak{m} + (x)$  on the top, giving a chain of length  $1 + \dim A$ . Thus  $\dim A[x] \geq 1 + \dim A$ .

Consider any chain of prime ideals in  $A[x]$ . By contracting to  $A$ , we get a chain of prime ideals in  $A$ . Between any two consecutive prime ideals in  $A$ , there can be at most two prime ideals in  $A[x]$ : The fiber over  $\mathfrak{p} \in \text{Spec } A$  is homeomorphic to the spectrum of  $k(\mathfrak{p}) \otimes A[x] = k(\mathfrak{p})[x]$  by Exercise 28, which has dimension 1. Thus the length of the chain in  $A[x]$  is at most  $1 + 2 \dim A$ .  $\square$

**Exercise 45** ([AM69]-exr-11.7). Let  $A$  be a Noetherian ring, then  $\dim A[x] = 1 + \dim A$ .

*Proof.* Say  $\mathfrak{p}$  is a prime ideal of  $A$  of height  $m$ . Localizing at  $\mathfrak{p}$  gives a Noetherian local ring  $A_{\mathfrak{p}}$  of dimension  $m$ . By the characterization of dimension [AM69, theorem 11.14], there exists an ideal  $\mathfrak{a} = (a_1, \dots, a_m)$  with  $\mathfrak{p}$  as one of its minimal prime ideals.

Now consider the height of  $\mathfrak{p}[x]$ . By Exercise 32, it is still a minimal associated prime ideal of  $\mathfrak{a}[x]$ , the latter is still generated by  $m$  elements. Thus by [AM69, corollary 11.16], the height of  $\mathfrak{p}[x]$  is at most  $m$ .

We also have that the height of  $\mathfrak{p}[x]$  is at least  $m$ , since every chain of prime ideals in  $A$  can be lifted to a chain of prime ideals in  $A[x]$  via the  $\mathfrak{p}_i \mapsto \mathfrak{p}_i[x]$ . So the height of  $\mathfrak{p}[x]$  is exactly  $m$ .

Now, for any prime ideal  $\mathfrak{q}$  of  $A[x]$  whose contraction to  $A$  is  $\mathfrak{p}$ , since  $\mathfrak{p}[x]$  is of height  $m$  and also contracts to  $\mathfrak{p}$ , by the same fiber argument in Exercise 44, the height of  $\mathfrak{q}$  is at most  $m + 1$ . Thus the dimension of  $A[x]$  is at most  $1 + \dim A$ . Combining with the other direction of inequality in Exercise 44, we have  $\dim A[x] = 1 + \dim A$ .  $\square$

## 8 Extra

**Exercise 46.** Let  $R := k[x_1, \dots, x_n]$ ,  $I := (x_1, \dots, x_n)$  is an  $R$ -ideal. Show that

$$R(I) \cong k[x_1, \dots, x_n, y_1, \dots, y_n] / (x_i y_j - x_j y_i)$$

where  $R(I)$  is the Rees algebra of  $I$ , that is,

$$R(I) := \bigoplus_{d=0}^{+\infty} I^d t^d \subseteq R[t]$$

*Proof.* Define a homomorphism

$$\varphi : k[x_1, \dots, x_n, y_1, \dots, y_n] \rightarrow R(I), \quad x_i \mapsto x_i, y_i \mapsto x_i t$$

It's clearly surjective. Note that

$$\varphi(x_i y_j - x_j y_i) = x_i x_j t - x_j x_i t = 0$$

Hence the ideal  $(x_i y_j - x_j y_i)$  lies in the kernel of  $\varphi$ . It remains to show that the kernel is exactly this ideal.

If some polynomial  $f(x_1, \dots, x_n, y_1, \dots, y_n)$  lies in the kernel, then  $f(x_1, \dots, x_n, x_1 t, \dots, x_n t) = 0$  in  $R(I)$ . Say  $f$

$$f(x_1, \dots, x_n, y_1, \dots, y_n) = \sum_{i_1, \dots, i_n, j_1, \dots, j_n} a_{i_1, \dots, i_n, j_1, \dots, j_n} x_1^{i_1} \dots x_n^{i_n} y_1^{j_1} \dots y_n^{j_n}$$

$f(x_1, \dots, x_n, x_1 t, \dots, x_n t) = 0$  shows that

$$\sum_{i_1, \dots, i_n; \sum j_{\bullet} = d} a_{i_1, \dots, i_n, j_1, \dots, j_n} x_1^{i_1 + j_1} \dots x_n^{i_n + j_n} = 0 \quad (1)$$

and in  $k[x_1, \dots, x_n, y_1, \dots, y_n] / (x_i y_j - x_j y_i)$ , note that

$$x_1^{i_1} \dots x_n^{i_n} y_1^{j_1} \dots y_n^{j_n} = x_1^{i'_1} \dots x_n^{i'_n} y_1^{j'_1} \dots y_n^{j'_n}$$

when  $i_k + j_k = i'_k + j'_k$  for all  $1 \leq k \leq n$ . Thus collecting terms with the same  $i_k + j_k$ , and apply Equation 1,  $f = 0$  in  $k[x_1, \dots, x_n, y_1, \dots, y_n] / (x_i y_j - x_j y_i)$ .  $\square$

## References

- [AM69] Atiyah, M. F. and Macdonald, I. G. *Introduction To Commutative Algebra*. Addison-Wesley Publishing Company, Inc., 1969.